

Umbilical Panel Mating Automation

BLUE ORIGIN

Team Members: McKenna McMurray (S-TLS), Will Sedo (S-TLF), Wilson Zambrano (S), Sheridan Dorsey (S), Elena Schwartz (J), Tay Han (J)

Faculty Advisor: Professor Gordon Krauss

Liaisons: Mark Abrams, Angela Hsueh ('04), Scott Fletcher



Source: SatNews



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Source: SatNews

Blue Origin & New Glenn



Source: SatNews

Blue Origin:

Aerospace company founded by Jeff Bezos, designs and manufactures spacecrafts and launch systems

Aim to lay the framework that will support easy and broad space accessibility: "Building a road to space"

Current Project:

New Glenn: Reusable first stage rocket to routinely carry large payloads to orbit

Upgrading rocket systems and mechanisms to increase launch frequency

Umbilical System



Source: SatNews

- Provides connections between the launch vehicle and ground support tower for the supply of fluid and electronics.
- Must be able to precisely align with the rocket under all environmental conditions of the pre-launch stage.
- Current mating process takes days and almost entirely manual, requiring workers to physically connect each panel and the tubing before manually verifying connection.

Privileged Information

As this project contains information pertaining to rocket technology, it falls under ITAR Part 121 – Category IV, which restricts distribution of certain information to “US persons.”

Portions of this project also include privileged Blue Origin information. Due to these restrictions and the open setting of this presentation, please understand that the information shared in this presentation is limited, and not all questions may be answered.

Presentation Agenda

- I. Blue Origin Background
- II. Problem Statement & Design Thinking
- III. Mechanical Testbed
 - A. Panel Design
 - B. Movement Mechanisms
- IV. Sensor Choices
- V. Control Algorithm
- VI. Next Steps

Problem Statement



Source: Space Visions Congress

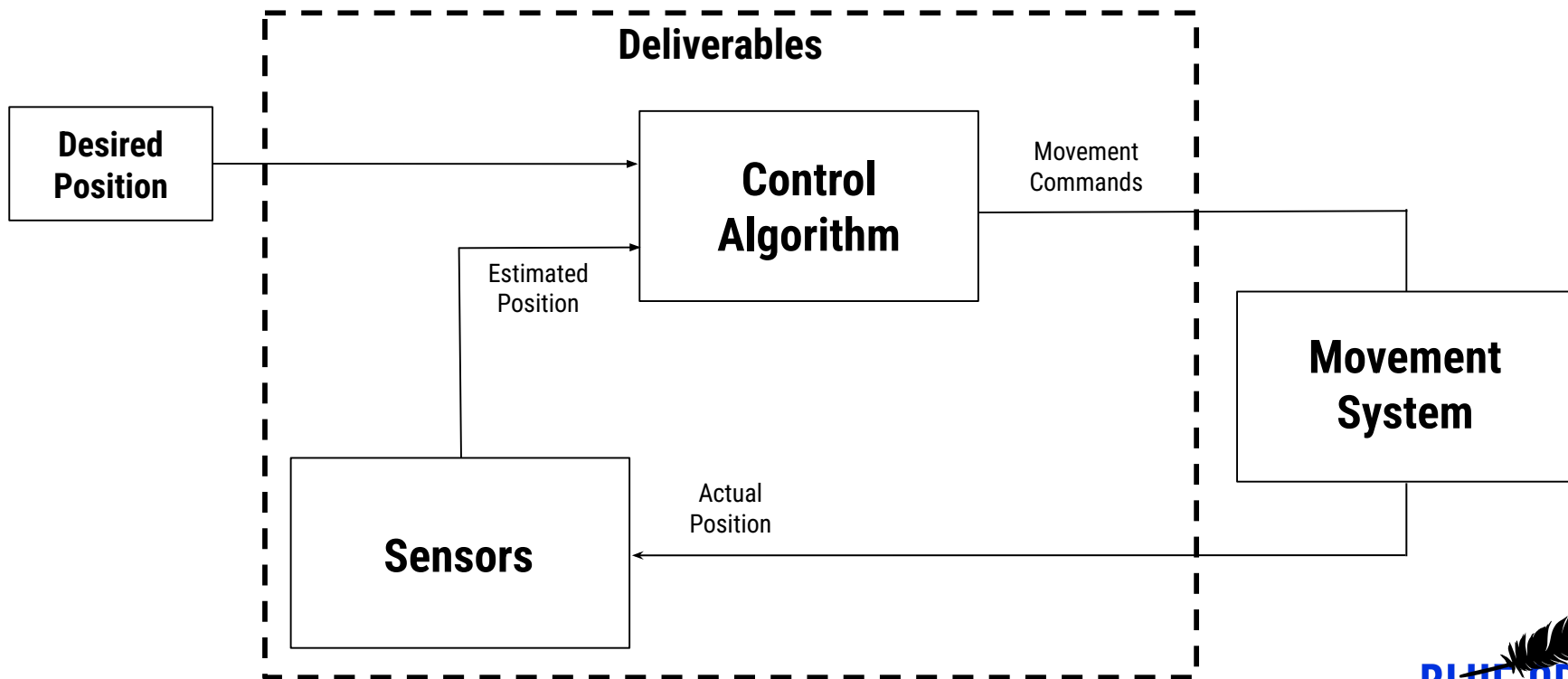
Blue Origin's current umbilical mating procedure for the New Glenn project does not meet future project goals, including concerns of alignment accuracy, human facilitation, and operational speed and efficiency. The Harvey Mudd College Blue Origin Clinic Team will evaluate sensor options and design a control algorithm and system to autonomously mate umbilical panels in the loading environment.

Project Impact

- Increases the rate of Blue Origin launches
 - Over 95% reduction in mating time if successful
 - Removes umbilical panel mating as bottleneck
- Decreases cost per launch for New Glenn
 - Each successful reuse avoids rebuilding a component
 - Saves money on production
 - Avoids additional emissions per launch
- Increases civilian access to space



Project Overview



Design Thinking

Functions	Objectives [metrics]	Constraints
• Senses vehicle location • Aligns panels autonomously • Corrects alignment error • Mates panels • Verifies continued connection	• Be time efficient [time to successful mate] • Be autonomous [# of tasks that must be completed by a human] • Be reusable [# of launches before maintenance] • Be refurbishable [cost of maintenance or replacement] • Utilize off-the-shelf parts [# of custom parts]	• No false contact • Withstands temperature range from space cold to launch hot • Can operate long-term in near-sea conditions including rain & fog • Costs < \$xx for xx launches • Completes mating in xx minutes

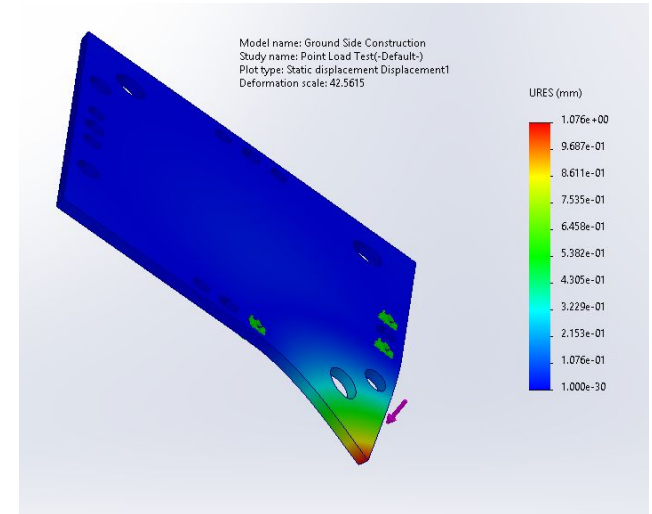
Mechanical Overview

To test sensors and algorithms, a physical testbed was required:

- Representative model of umbilical panels (both ground and vehicle sides)
- Controllable actuation for both panels

Project Panel Design + Manufacturing

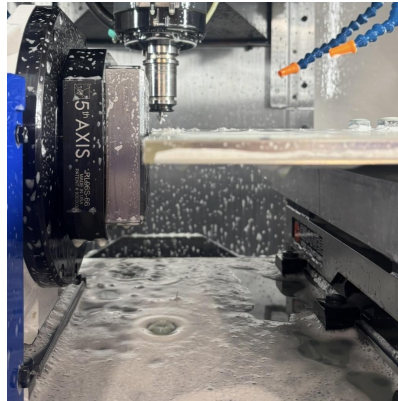
- Material Selection
 - 6061-T6 Aluminum
 - Thickness: 3/8"
- The panel must be designed to accommodate:
 - Mechanical alignment features
 - Integration points to the movement system
 - Mounting provisions for sensors
 - Specified connectors (6 total per panel)



With a 1000N point load, the 0.375" panel deflected only 1.07mm

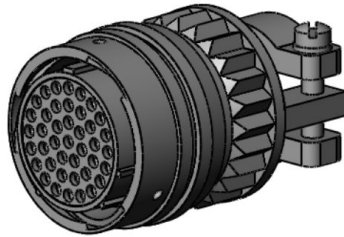
Panel Design Manufacturing

- Early prototyping was done on wood panels to inform the final design made on a 6061-T6 aluminum 3/8" panel.
- Final design was machined on a Brother Speedio 5 axis mill.

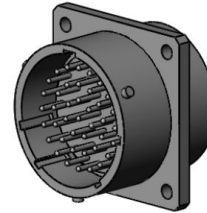


Connectors

- Connectors have tight tolerances
- Connectors are how connection will be assessed.
- Modifications were made to them



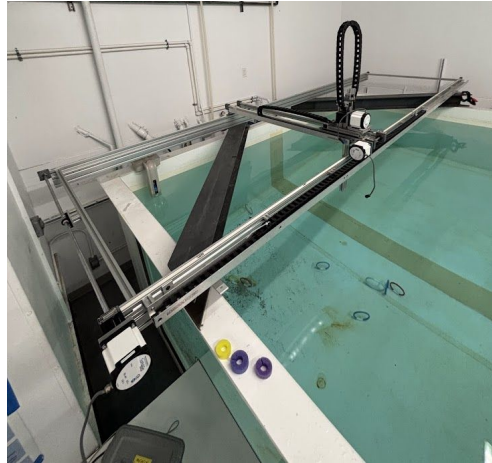
Ground Panel Connectors



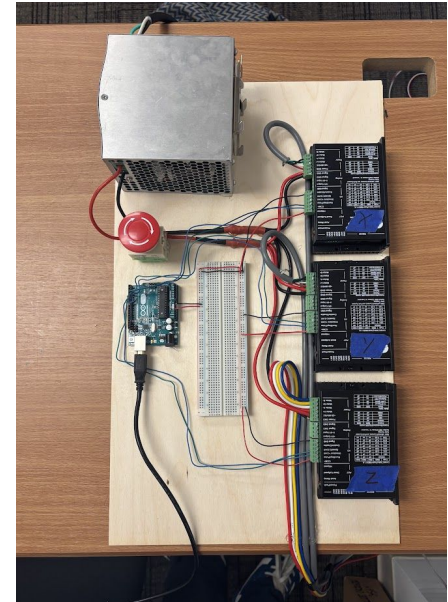
Vehicle Panel Connectors

Ground Panel Movement

- Requires active control in 4 axes
- Using existing 4m x 1m x 0.3m XYZ gantry with external tilting mechanism
- Arduino-controlled stepper motors



Gantry system used for ground panel movement



Gantry control electronics

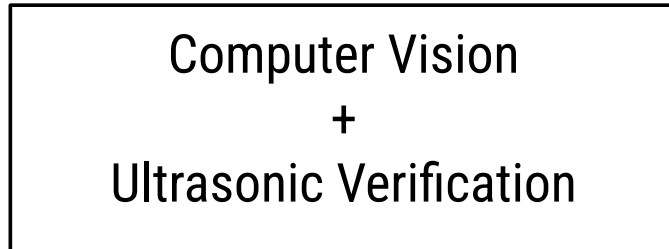
Vehicle Panel: Gantry System

- Requires passive movement in 3 DoF- 2 axes of translation and one rotation
- Mechanism based on T Bridge linear guide actuator setup
- Moves in oscillatory motion with additional disturbance noise to be representative of the environmental conditions.

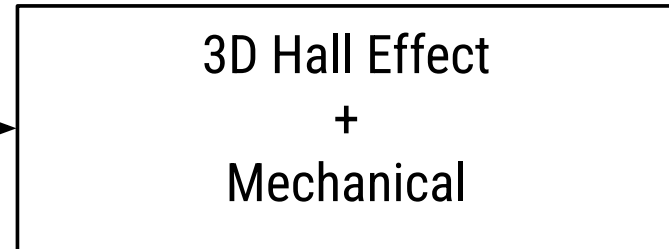


Controls & Sensors Overview

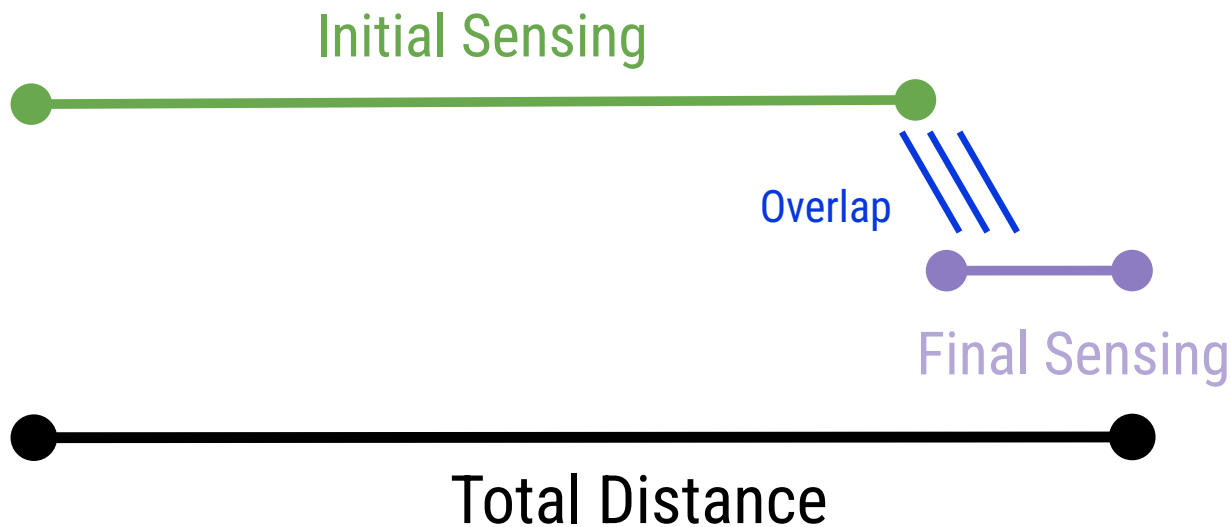
Initial Alignment &
Forward Movement



Final Alignment &
Closing Movement

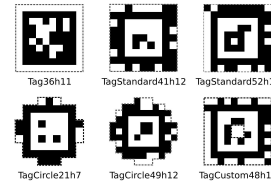


Controls & Sensors Overview

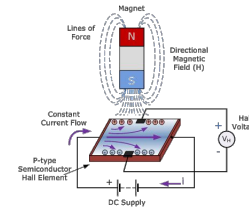


Sensor Development Focus

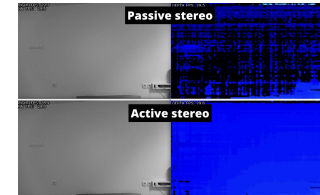
1. Refining distance-measurement accuracy across all selected sensors
2. Calibrating sensors under different environmental and operational conditions
3. Integrating sensor feedback into the control loop for smoother alignment and reduced steady-state error



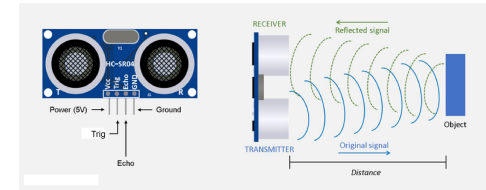
Source: APRIL robotics lab



Source: Electronics Tutorials



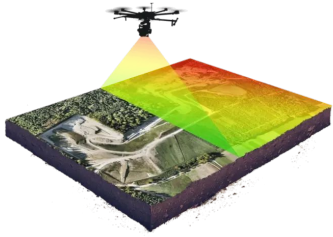
Source: Luxonis Docs



Source: Piolin2

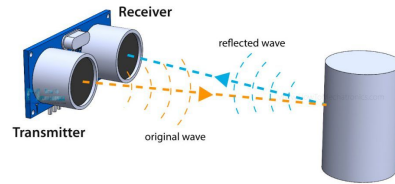
Sensor Team Focus

LiDAR



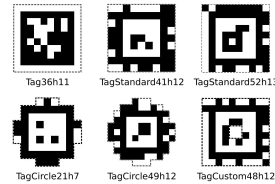
Source: FlyAbility

Ultrasonic



Source: Circuit Geeks

Computer Vision (CV)



Source: April Robotics Lab

Linear Potentiometer



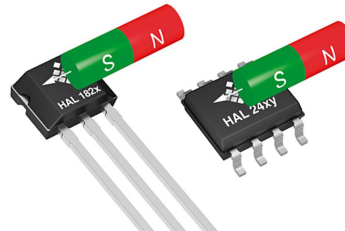
Source: Reichelt

Ultra Wideband Sensing (UWB)



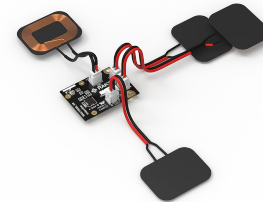
Source: STMicroelectronics

Hall Effect



Source: TDK-Micronas

Inductive Sensing



Source: RAK Wisblock

Initial Sensing



Initial Sensing



Total Distance

Initial movement will bring panels to close distance

Main sensor: Computer Vision (CV)

Secondary sensors

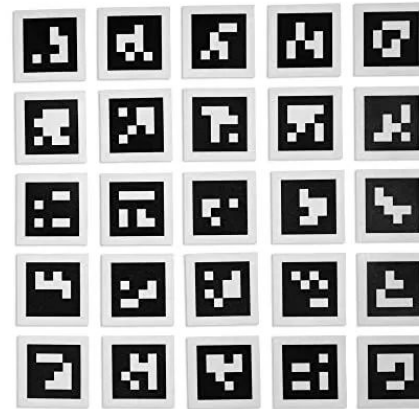
- Ultrasonic
- Stereo Vision
- Verification for CV

CV Fiducials

April Tags: unique 2D fiducial markers

- Visual barcodes for precise camera localization
- Many open-source libraries
- Standard RGB cameras

A single tag can capture translation and rotation



<https://m.media-amazon.com/images/I/51TTORyWL.jpg>



Source: Luxonis

CV Fiducials and Multitag

Multiple tags = more verification

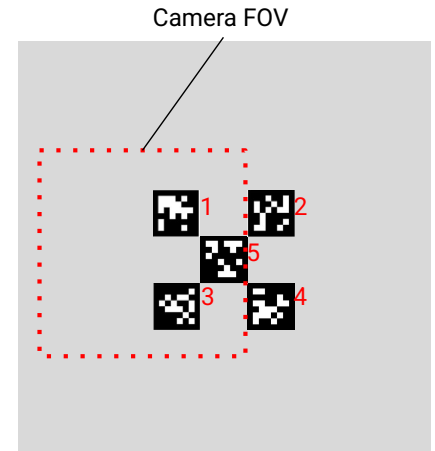
- Large to small (Russian nested dolls)

Perspective-n-Point (PnP)

- Estimating relative pose in 6 DoF given 3D-2D point correspondences
- OpenCV function solvePNP

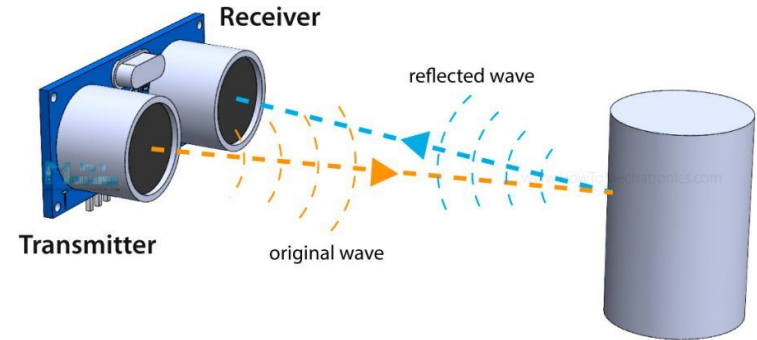


Source: Macalester College



Ultrasonic Sensing

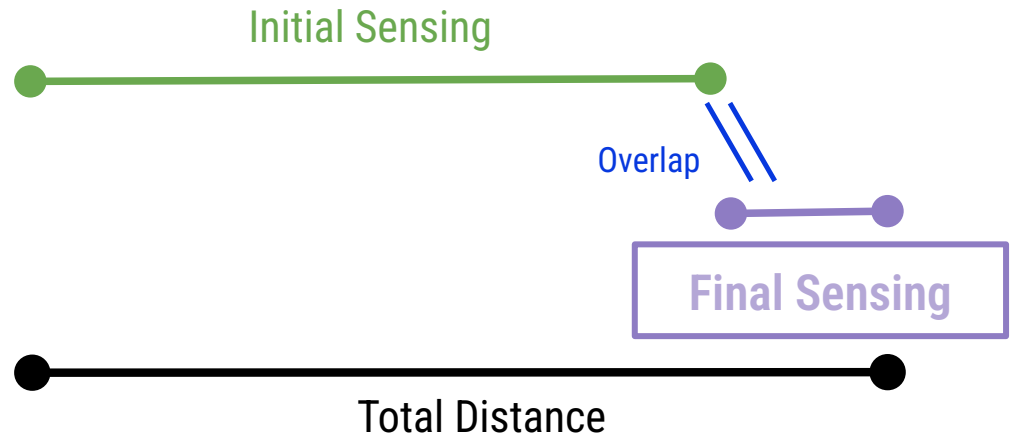
- Ultrasonic functions as Z translation verification for the CV system
 - Distance information from the camera is corroborated by the ultrasonic sensor
- Consistent accuracy through entire mating process



Source: Circuit Geeks

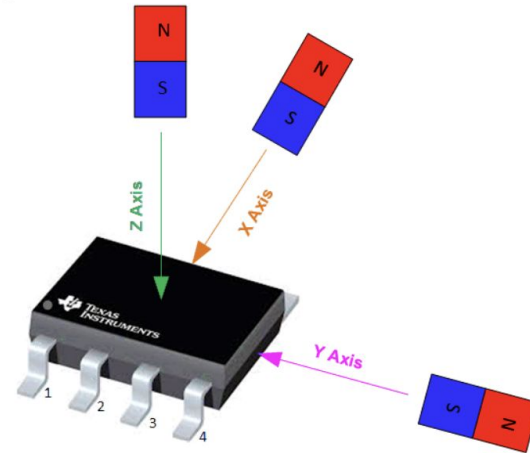
Final Sensing

- More precise measurements for final approach
 - Important for preventing false contact
- Selected Hall Effect sensor



Hall Effect Sensor

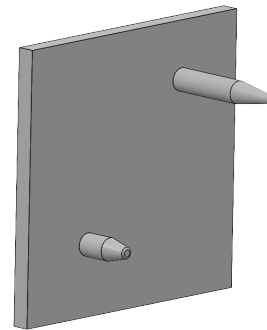
- Hall Effect Sensor: fixed current running through thin strip of metal (Hall plate)
- 3D Hall Effect Sensor: three orthogonal Hall plates to capture X, Y, Z magnetic readings and convert to distance
- Trained a machine learning model on hall effect data collected on a 3-axis mill to output expected distance from magnet



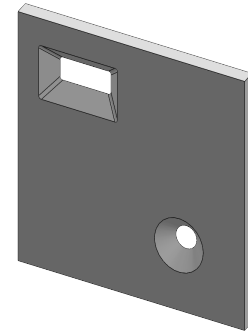
Source: TI TMAG5170

Alignment Slots and Pins

- Constrain one degree of freedom at a time
- X translation
 - Long alignment pin in slot
- Y translation
 - Short alignment pin in hole
- No rotation around X or Z axes



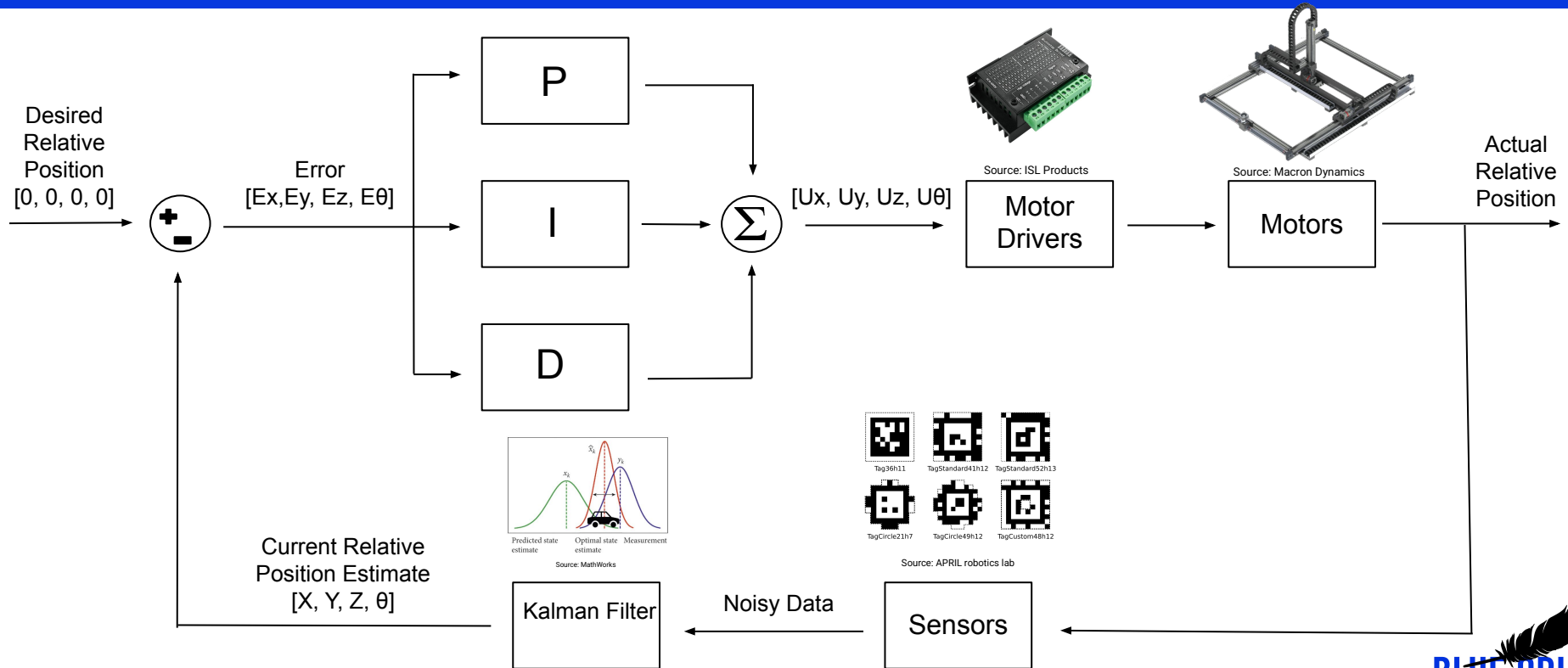
Ground Panel



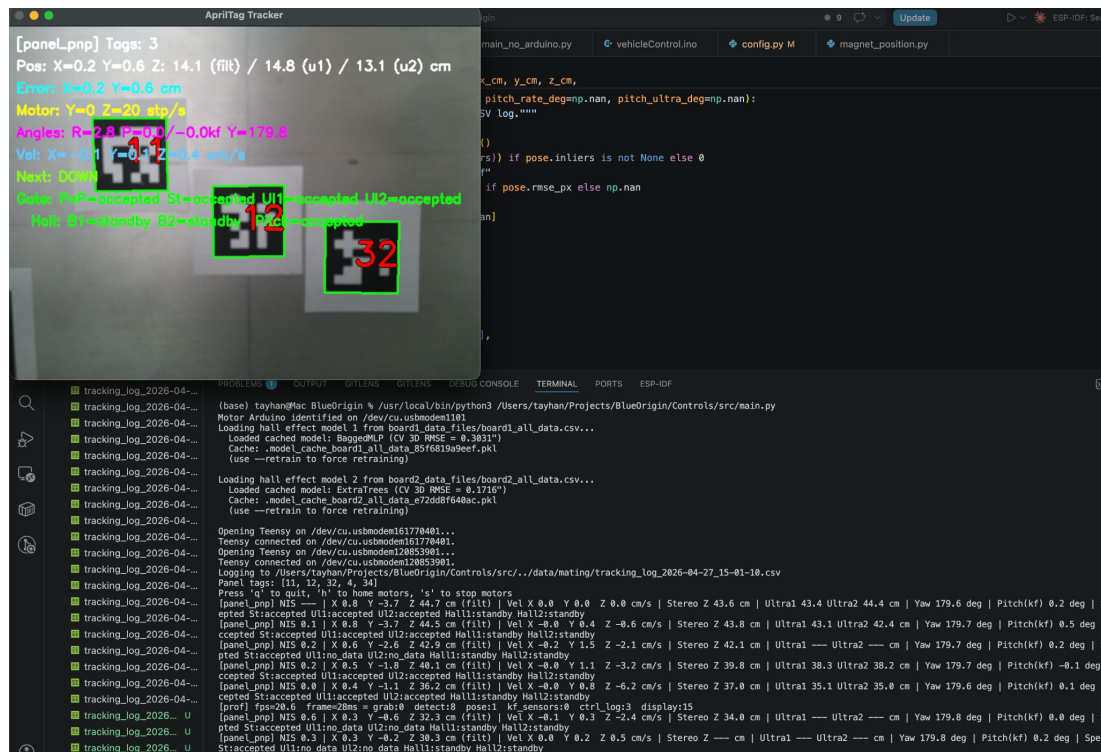
Vehicle Panel

CAD of Alignment Pin
Ideation

Control Algorithm Overview



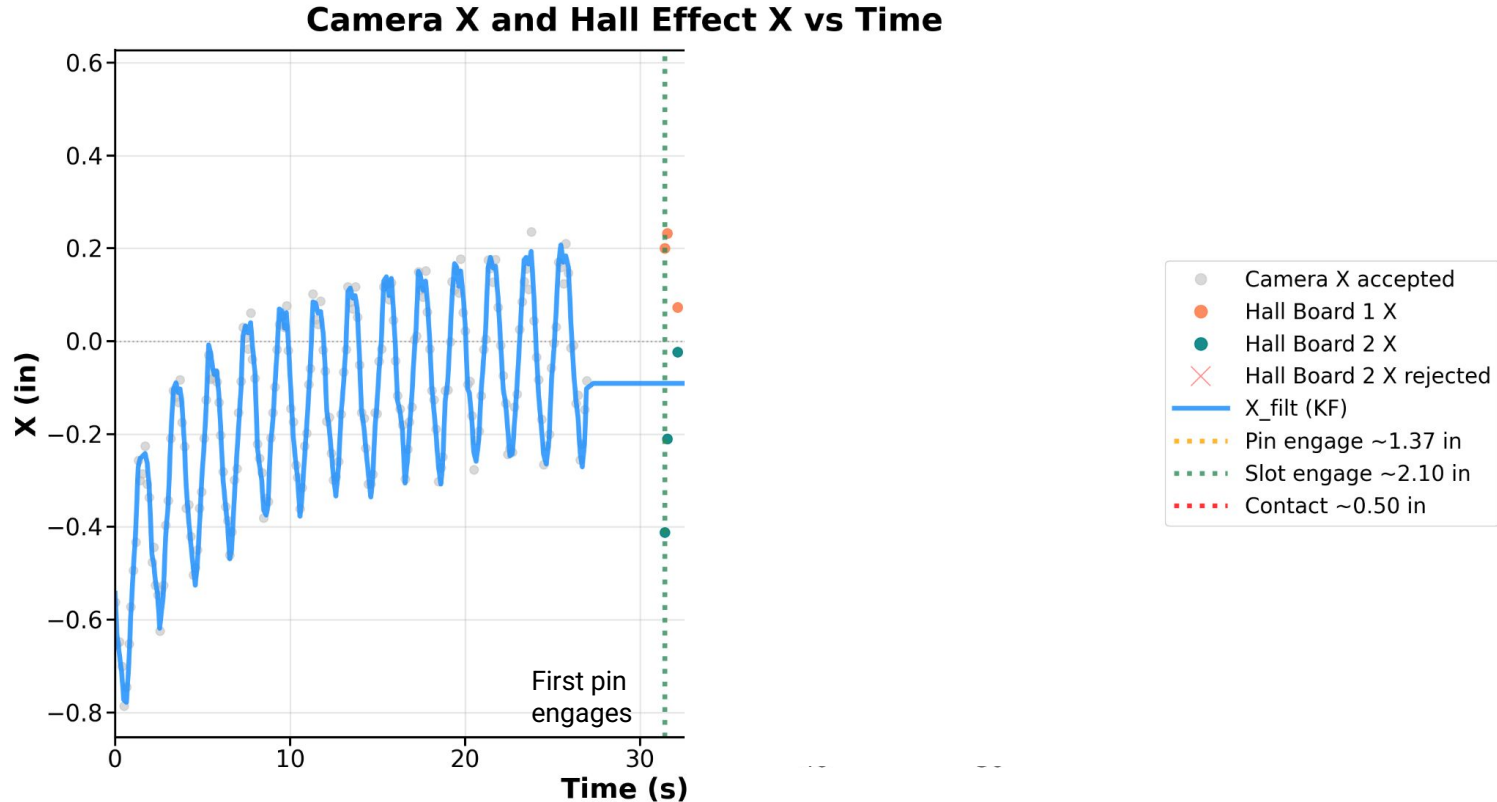
Testing



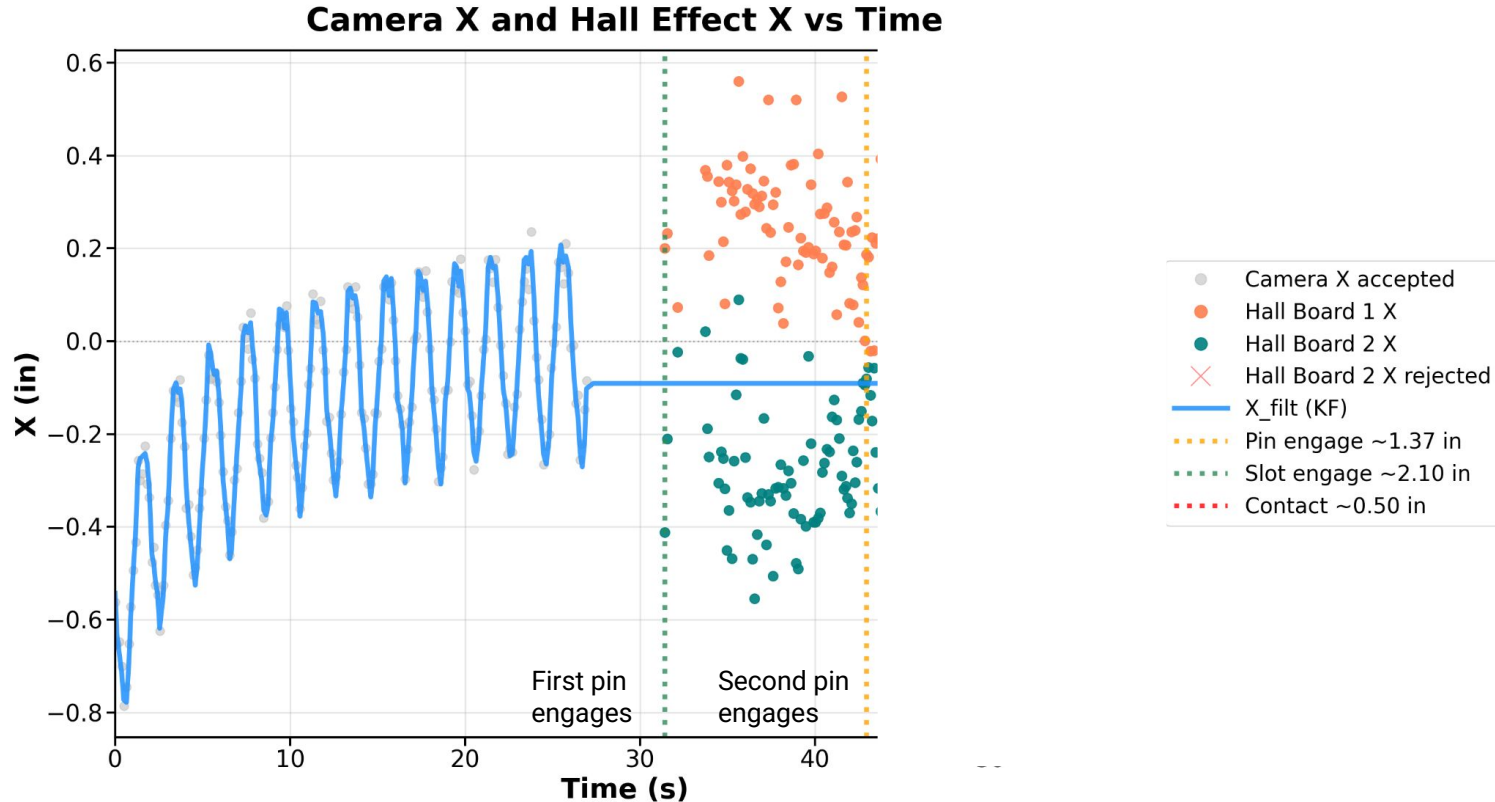
Procedure:

- Move panels to initial randomized position
- Run alignment procedure
- Diagnose with terminal and HUD overlay
- Verify CSV files

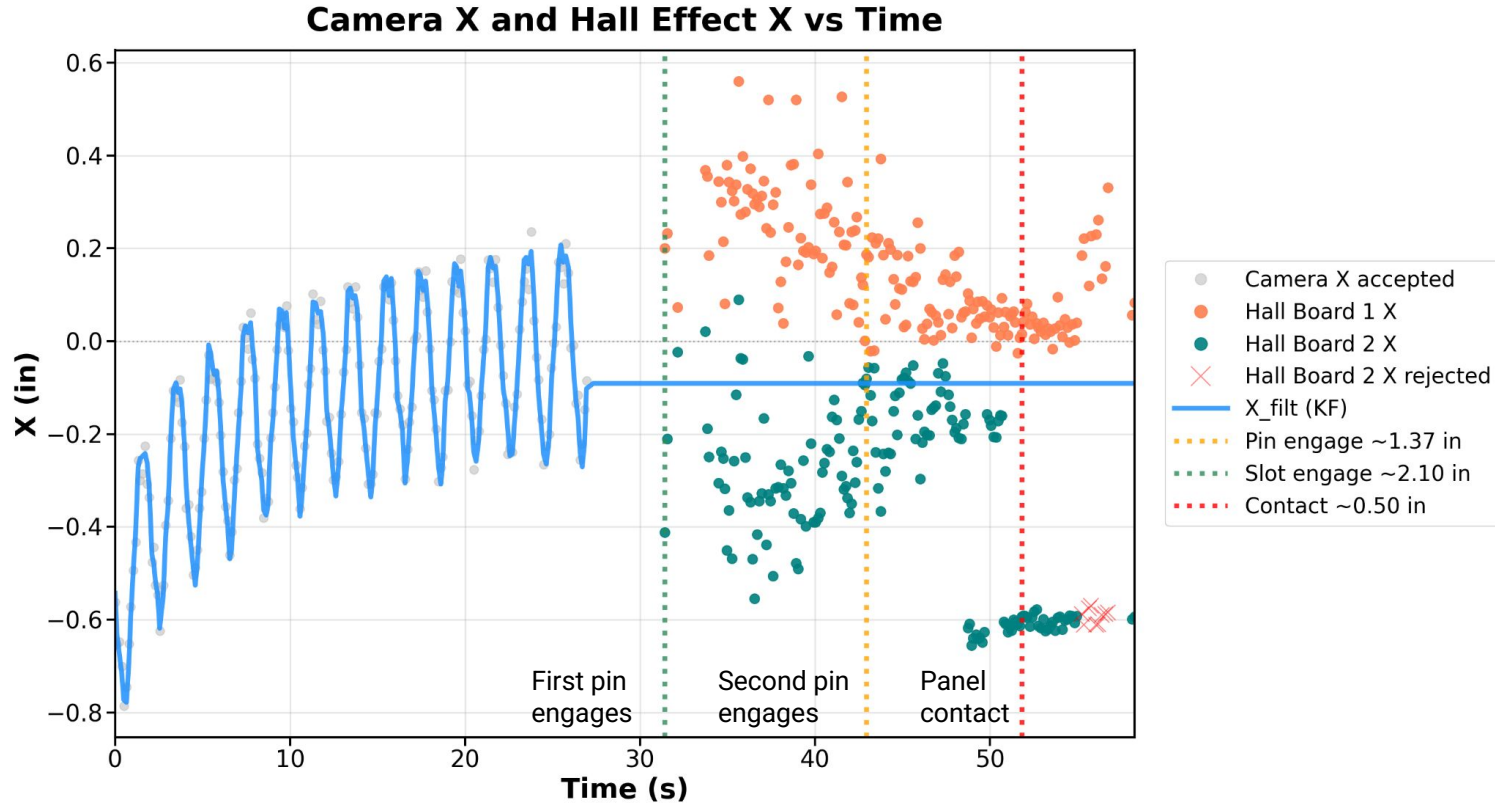
Standard Trial Example



Standard Trial Example

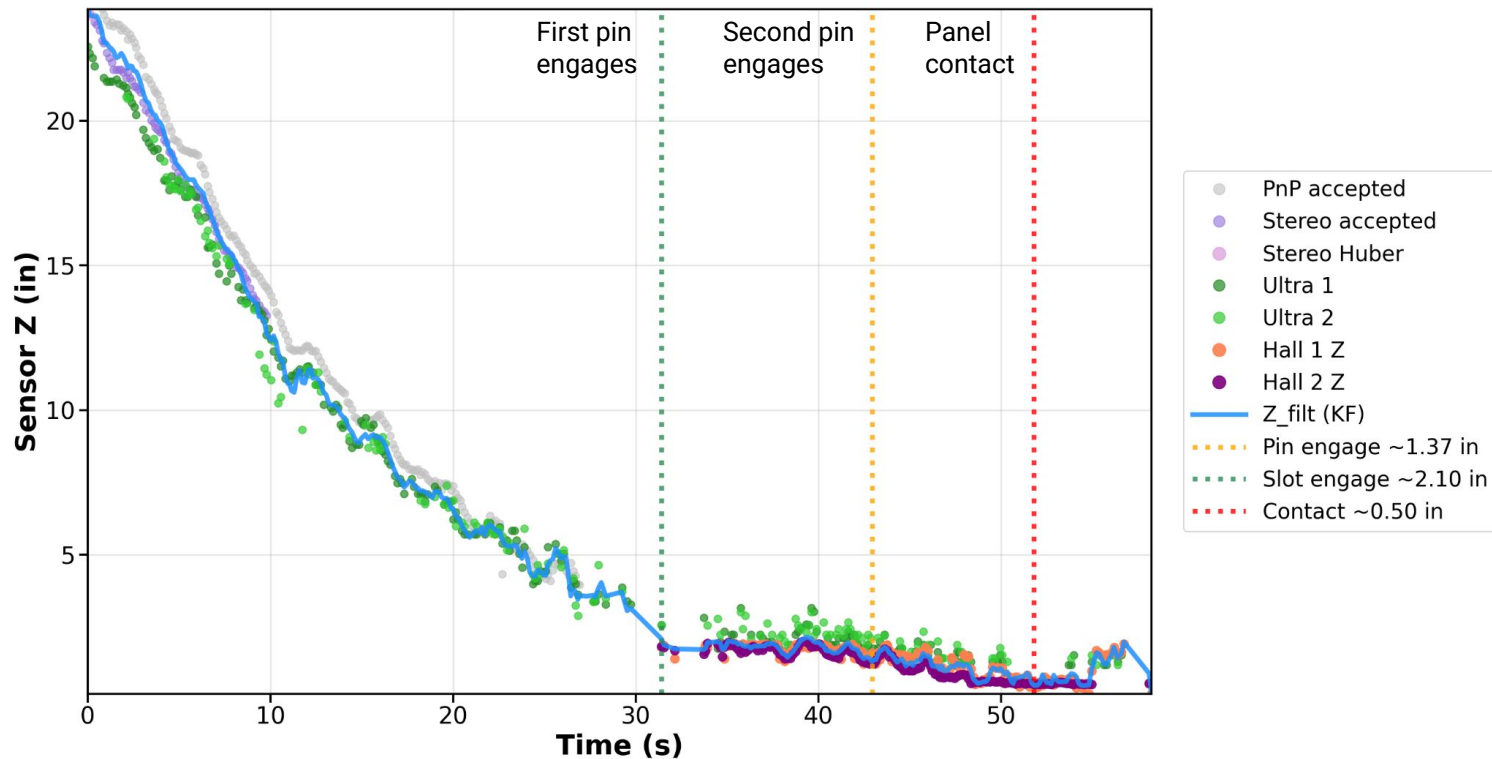


Standard Trial Example

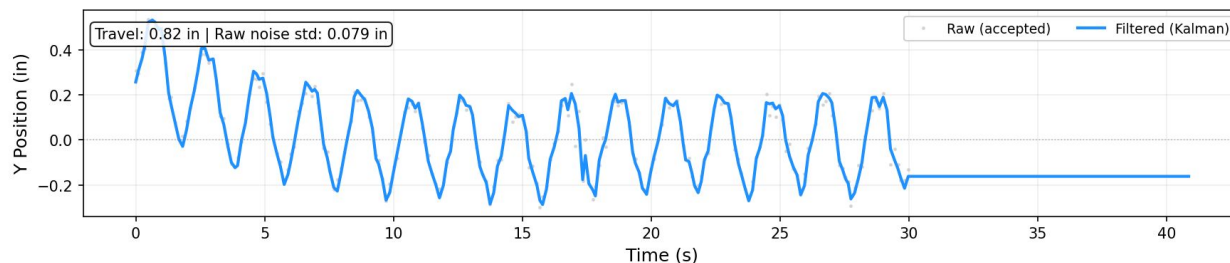


Standard Trial Example

Depth Comparison : All Z Sensors vs Time



Results



- Most significant modalities are Computer Vision and mechanical alignment pins
 - Verified through sensor degradation trials and additional impact trials

Future Work + Blue Origin Implementation

Next Steps for Project Development

- Expanding to 6 Degrees of Freedom
- Improving robustness of load-bearing modalities
- Implementing force feedback

Steps for Blue Origin Implementation

- Embedded system
- Stewart platform system instead of gantry system
- Implementing a system-specific control strategy (LQR)

Special Thank You

Fall Juniors: Eleanor Champlin-Wilson, Luke Vlases, Thiven Anderson

Engineering Faculty: Professor Krauss, Professor Spencer, Professor Mendelson, Professor Mohanty

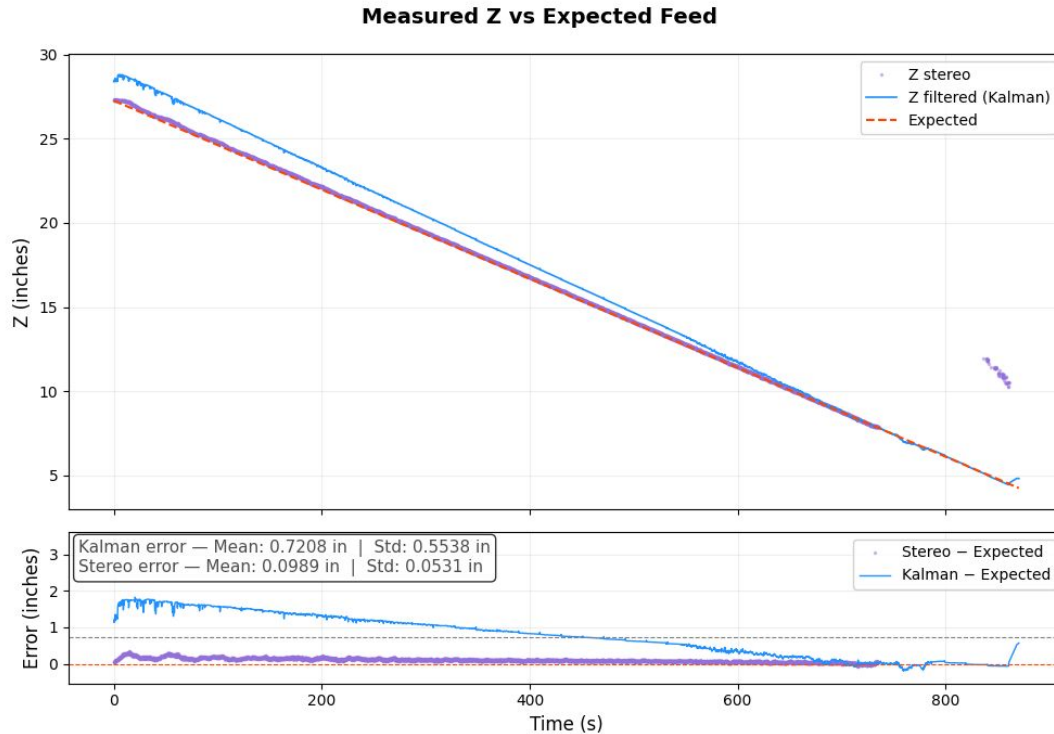
Engineering Staff: Lorena Gonzalez, Sydney Torrey, Drew Price, Xavier Walter, Jacob Stampiel

Liaisons: Mark Abrams, Angela Hsueh ('04), Scott Fletcher

Thank you!

Questions?

Multitag Accuracy: Moving Trials

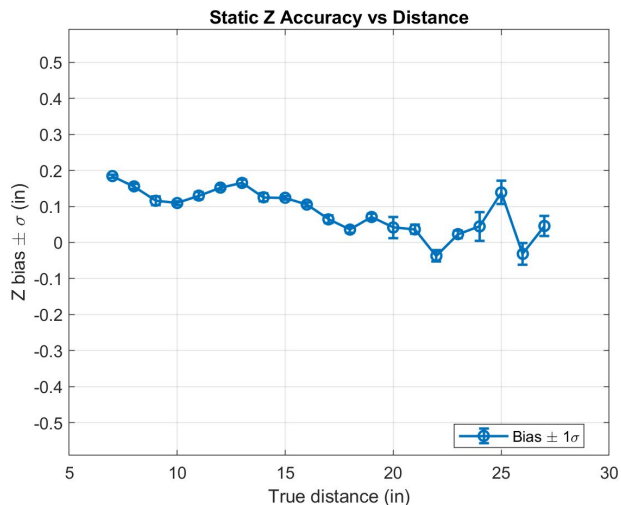


<0.01" Mean Stereo Vision Error

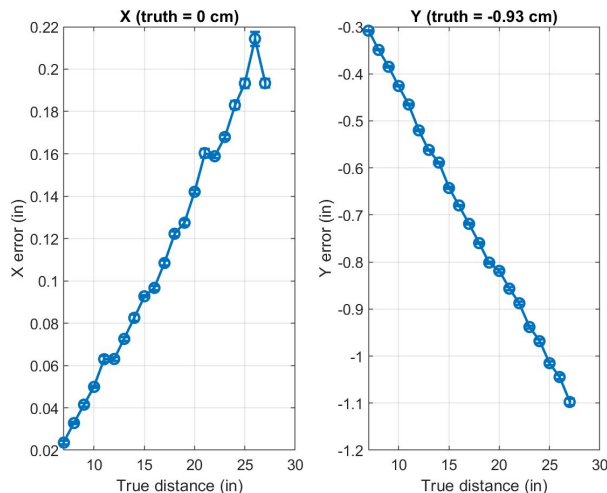
PnP sensitive to bias



Multitag Accuracy: Static Trials



Z-distance measurements maintain sub-5 mm bias with low scatter across all ranges.



Systematic lateral drift indicates an intrinsic calibration or alignment bias, likely due to non-zero angles in the experimental setup

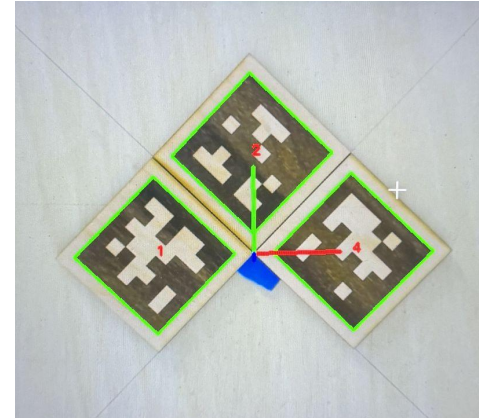
Computer Vision Design Selection

Computer Vision

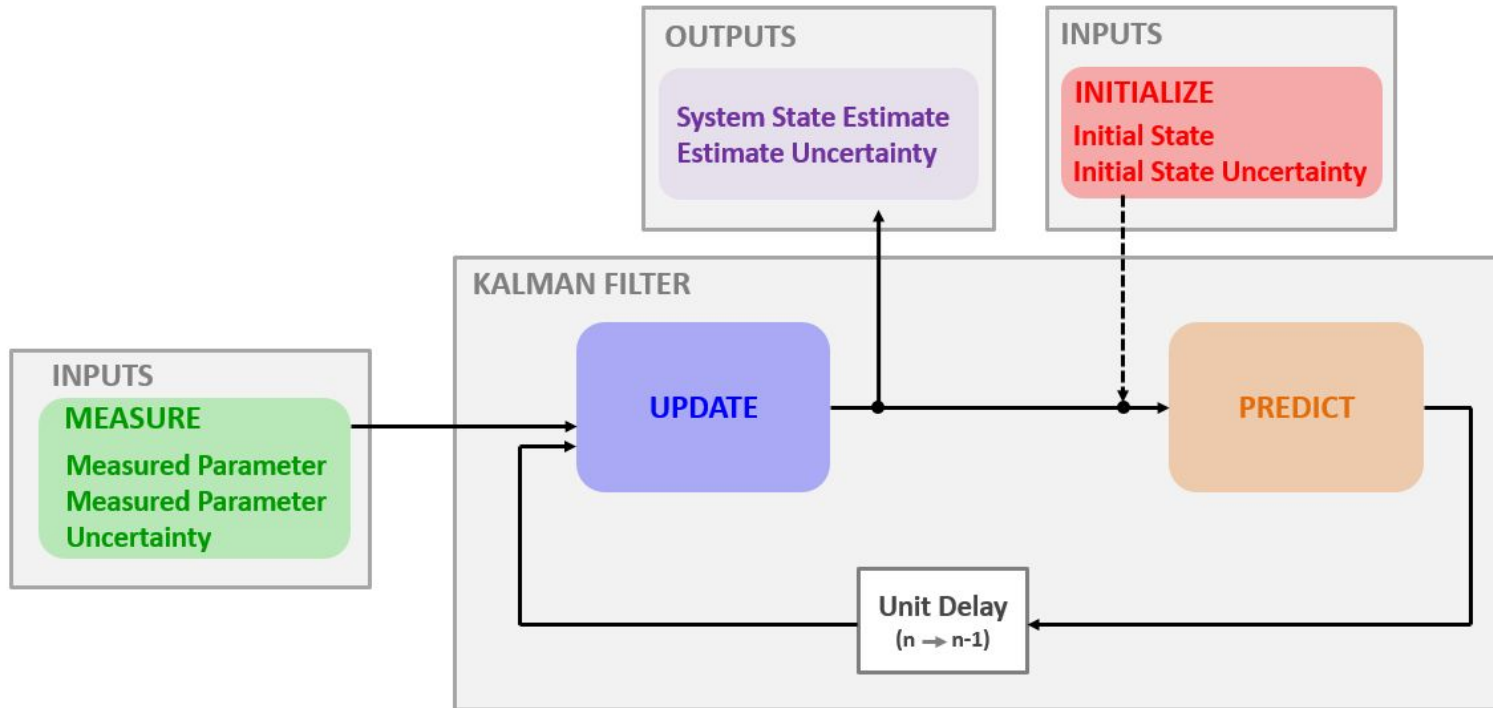
- Clutterage: can print fiducial markers directly on panels
- 6 DoF pose at close distances

LiDAR

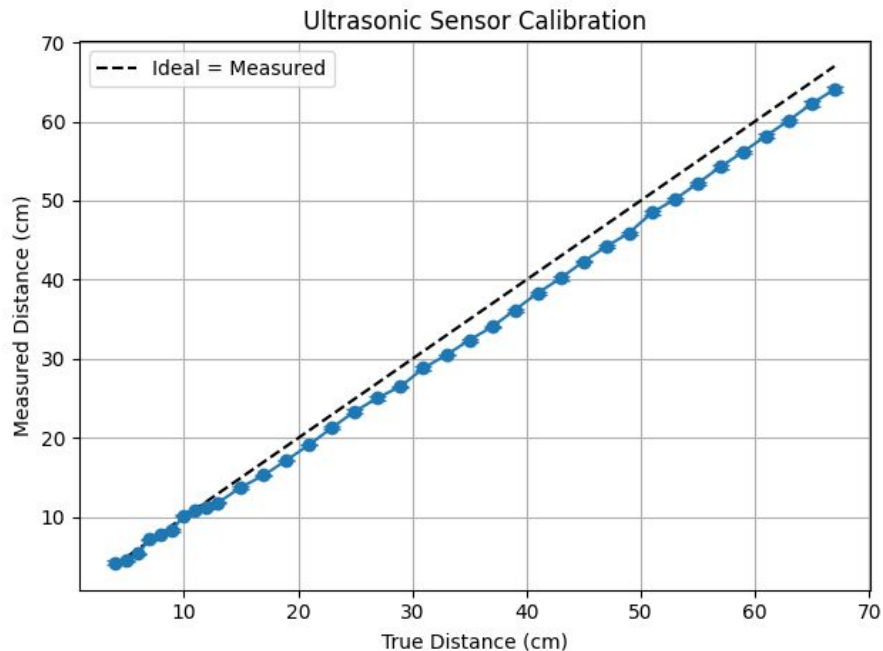
- Cost tradeoff for high-precision
- Environmental conditions for laser



Kalman Filter Explanation



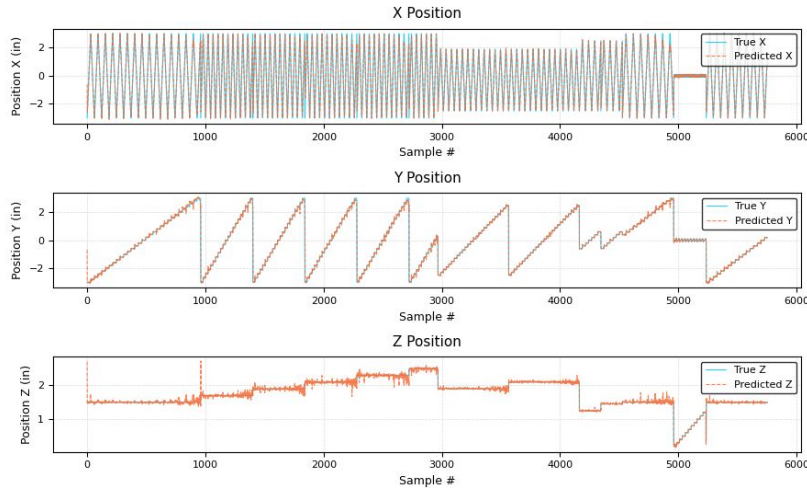
Ultrasonic Testing



Preliminary testing of ultrasonic sensor shows
slight drift at larger distances

Hall Effect Sensor Data

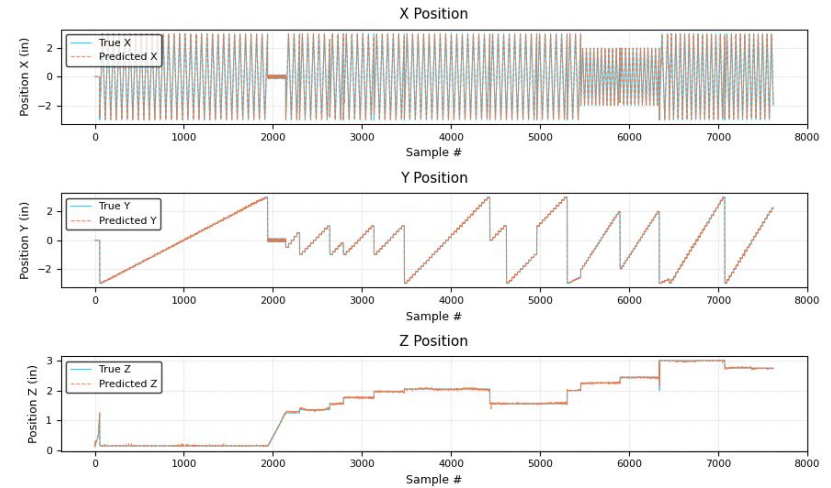
Board1 All Data — True Position vs. Model Predicted Position



Board 1 Hall Effect Model

RMSE: 0.2994 inches

Board2 All Data — True Position vs. Model Predicted Position



Board 2 Hall Effect Model

RMSE: 0.1716 inches